

BIOLOGY

TENTH EDITION

Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson

15

Linkage and Chromosomes



Lecture Presentation by
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Kathleen Fitzpatrick

Locating Genes Along Chromosomes

- a) Mendel's "hereditary factors" were purely abstract when first proposed
- b) Today we can show that the factors—genes—are located on chromosomes
- c) The location of a particular gene can be seen by tagging isolated chromosomes with a fluorescent dye that highlights the gene

Figure 15.1

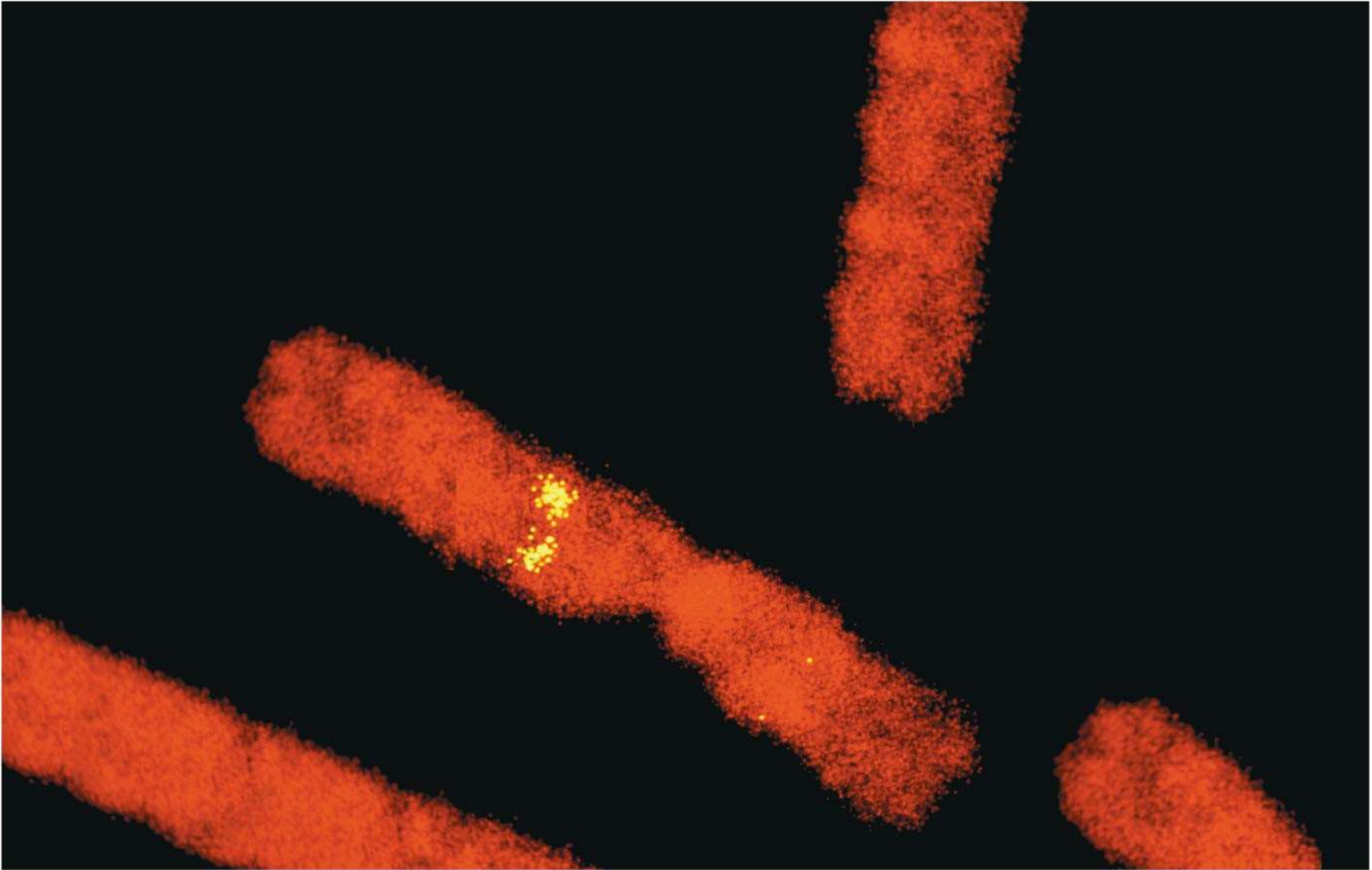
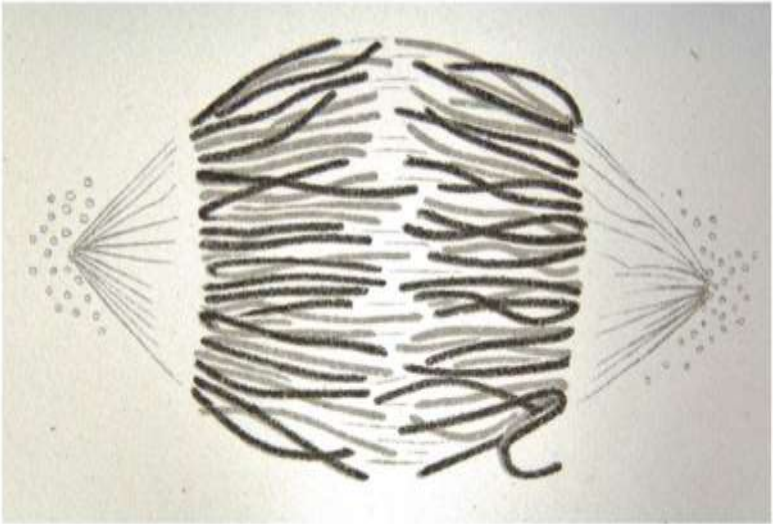


Figure 15.1a



- a) Cytologists worked out the process of mitosis in 1875, using improved techniques of microscopy
- b) Biologists began to see parallels between the behavior of Mendel's proposed hereditary factors and chromosomes
- c) Around 1902, Sutton and Boveri and others independently noted the parallels and the **chromosome theory of inheritance** began to form

Figure 15.2

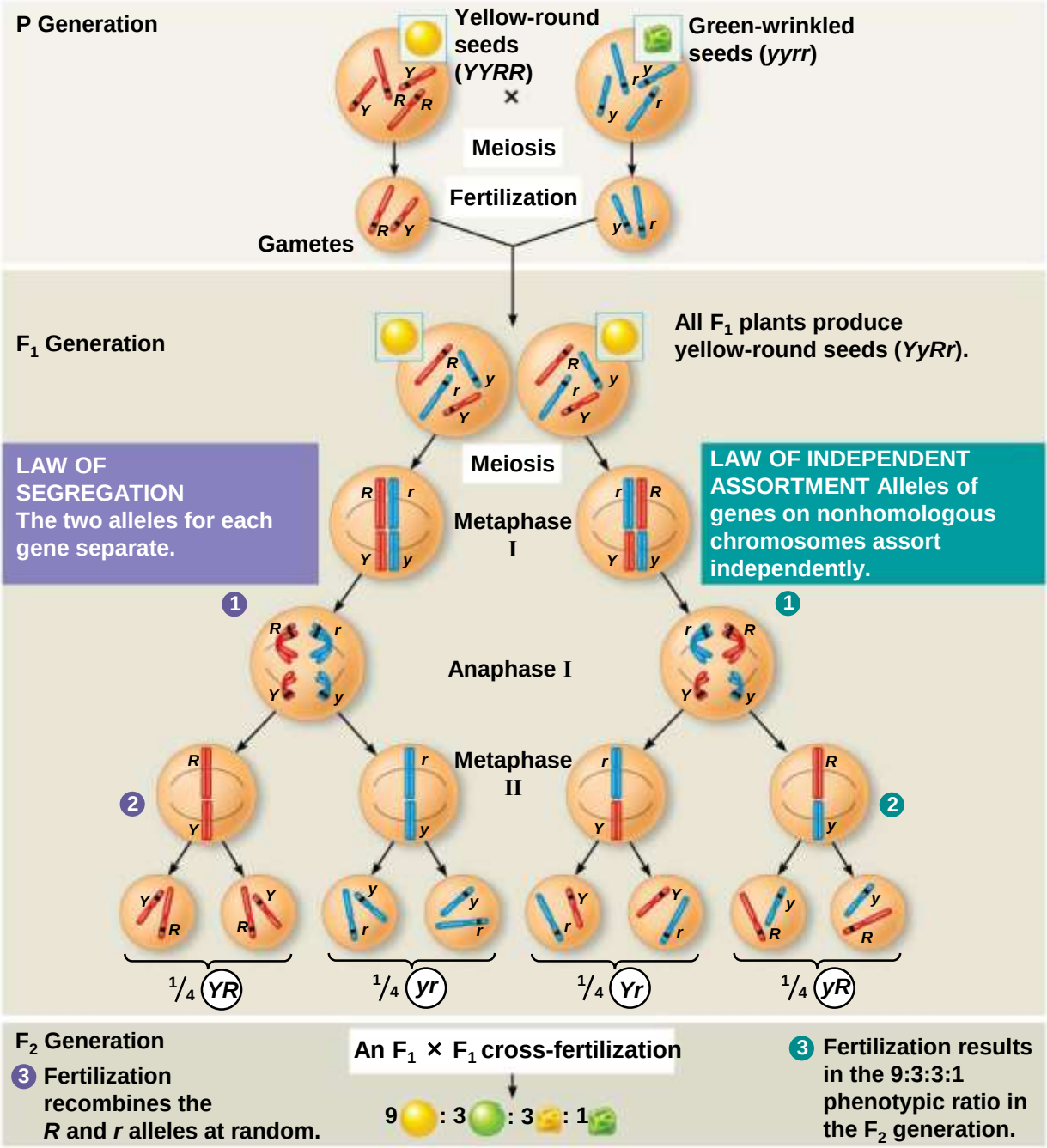


Figure 15.2a

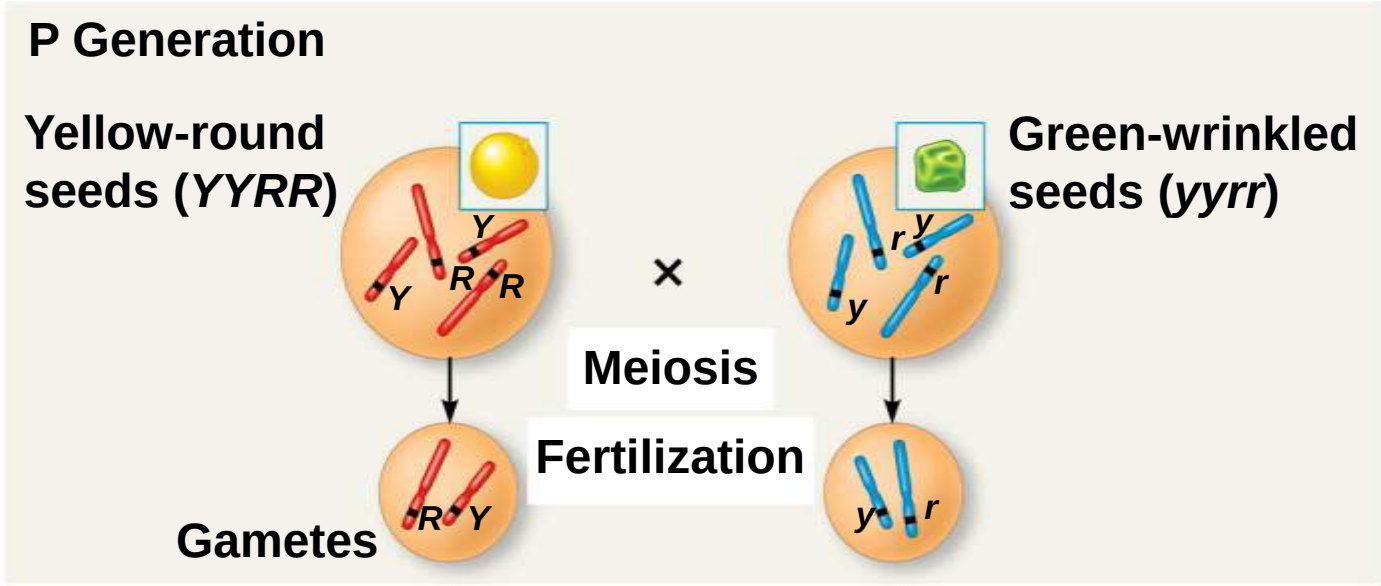
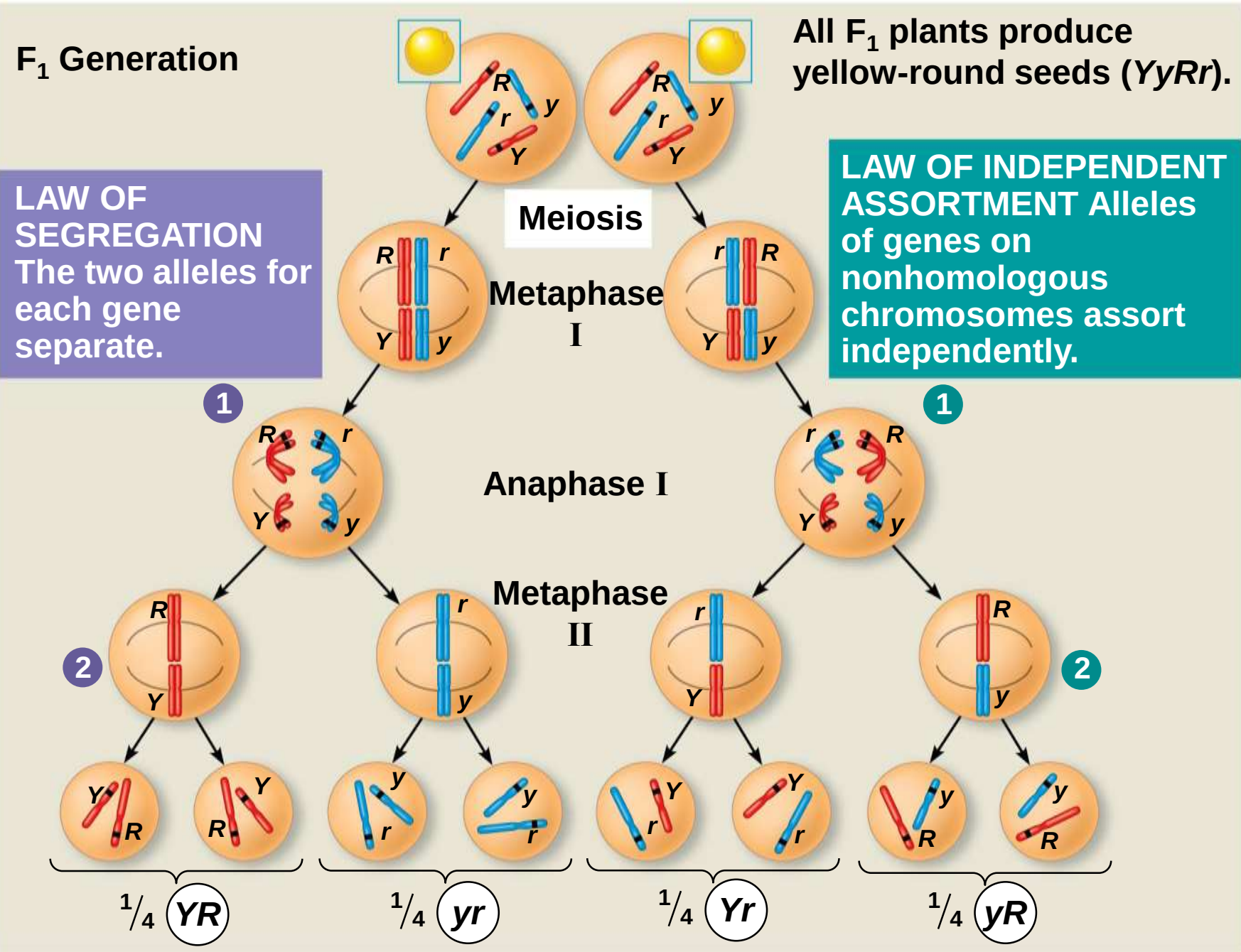


Figure 15.2b



LAW OF SEGREGATION

F₂ Generation

- ③ Fertilization recombines the *R* and *r* alleles at random.

An F₁ × F₁ cross-fertilization



LAW OF INDEPENDENT ASSORTMENT

- ③ Fertilization results in the 9:3:3:1 phenotypic ratio in the F₂ generation.

Concept 15.1: Morgan showed that Mendelian inheritance has its physical basis in the behavior of chromosomes: *Scientific inquiry*

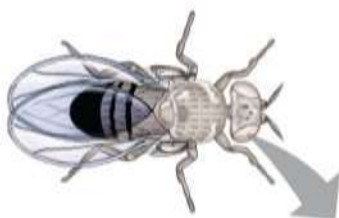
- a) The first solid evidence associating a specific gene with a specific chromosome came in the early 20th century from the work of Thomas Hunt Morgan
- b) These early experiments provided convincing evidence that the chromosomes are the location of Mendel's heritable factors

Morgan's Choice of Experimental Organism

- a) For his work, Morgan chose to study *Drosophila melanogaster*, a common species of fruit fly
- b) Several characteristics make fruit flies a convenient organism for genetic studies
 - a) They produce many offspring
 - b) A generation can be bred every two weeks
 - c) They have only four pairs of chromosomes

- a) Morgan noted **wild type**, or normal, phenotypes that were common in the fly populations
- b) Traits alternative to the wild type are called mutant phenotypes

Figure 15.3

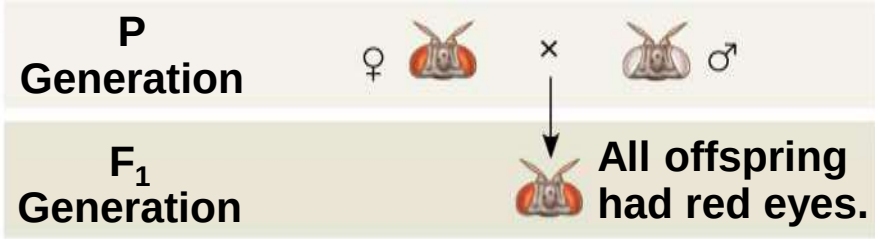


Correlating Behavior of a Gene's Alleles with Behavior of a Chromosome Pair

- a) In one experiment, Morgan mated male flies with white eyes (mutant) with female flies with red eyes (wild type)
 - a) The F_1 generation all had red eyes
 - b) The F_2 generation showed a 3:1 red to white eye ratio, but only males had white eyes
- b) Morgan determined that the white-eyed mutant allele must be located on the X chromosome
- c) Morgan's finding supported the chromosome theory of inheritance

Figure 15.4

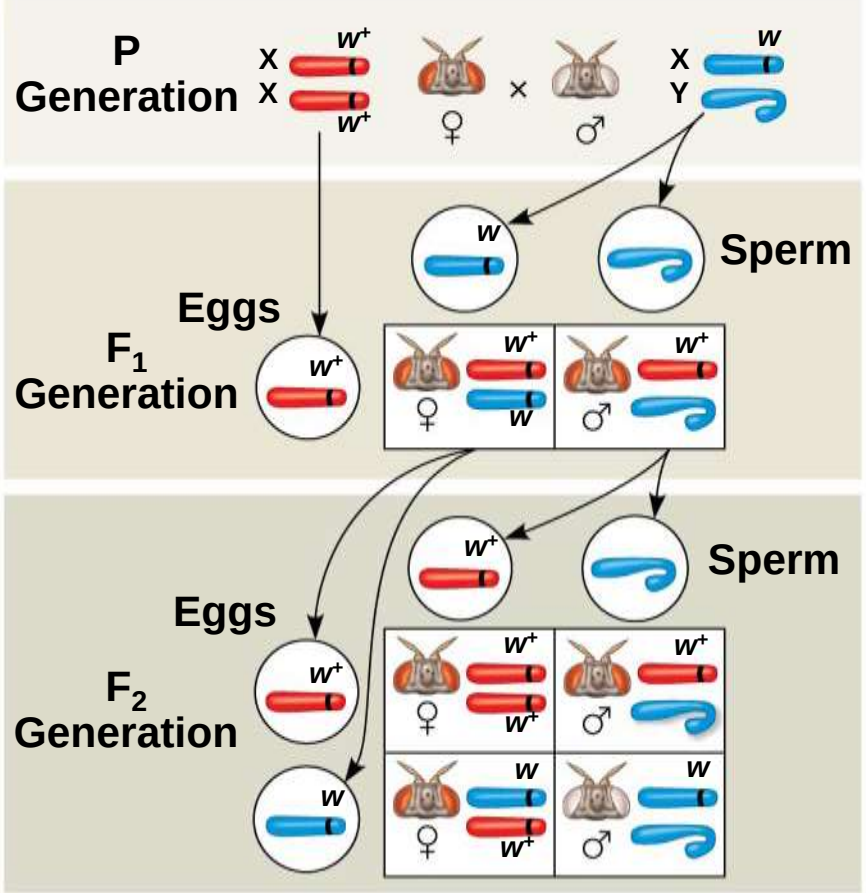
Experiment



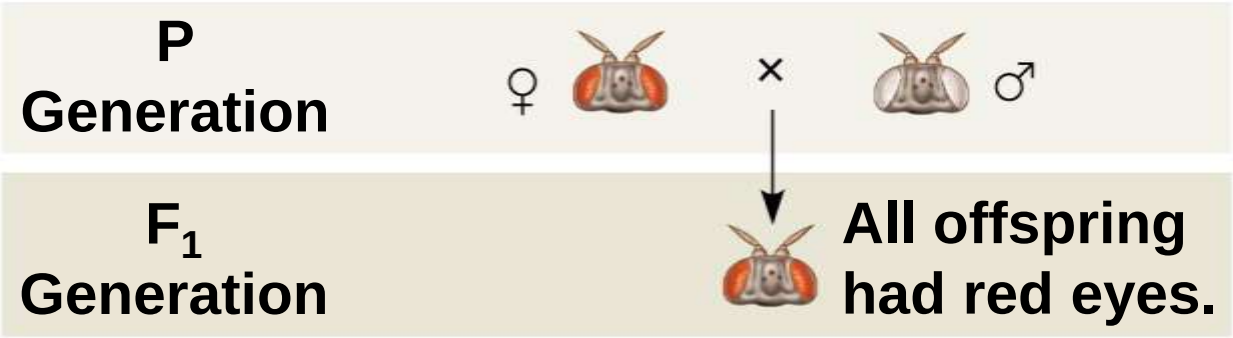
Results



Conclusion



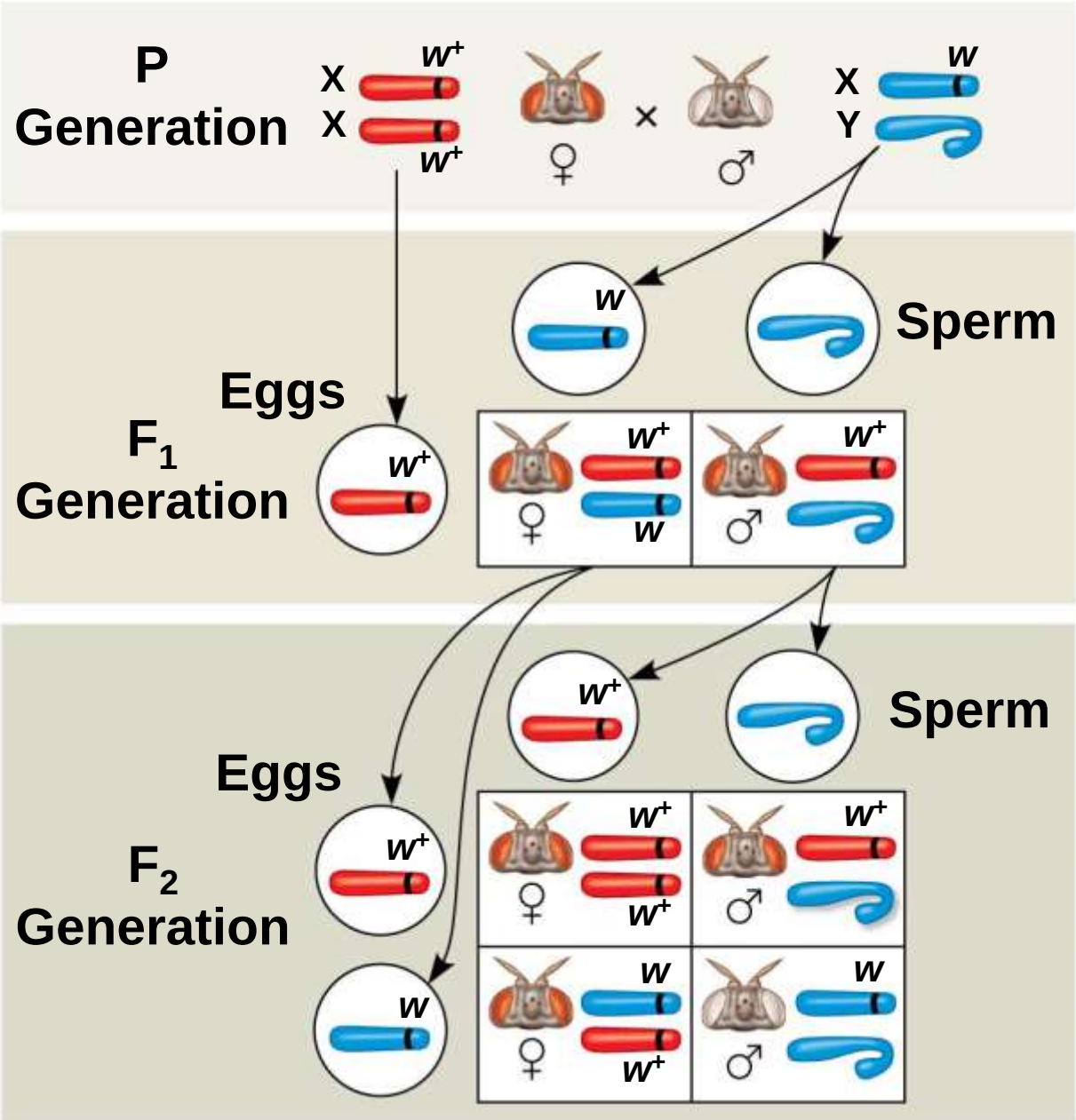
Experiment



Results



Conclusion



Concept 15.2: Sex-linked genes exhibit unique patterns of inheritance

- a) Morgan's discovery of a trait that correlated with the sex of flies was key to the development of the chromosome theory of inheritance
- b) In humans and some other animals, there is a chromosomal basis of sex determination

The Chromosomal Basis of Sex

- a) In humans and other mammals, there are two varieties of sex chromosomes: a larger X chromosome and a smaller Y chromosome
- b) A person with two X chromosomes develops as a female, while a male develops from a zygote with one X and one Y
- c) Only the ends of the Y chromosome have regions that are homologous with corresponding regions of the X chromosome

Figure 15.5

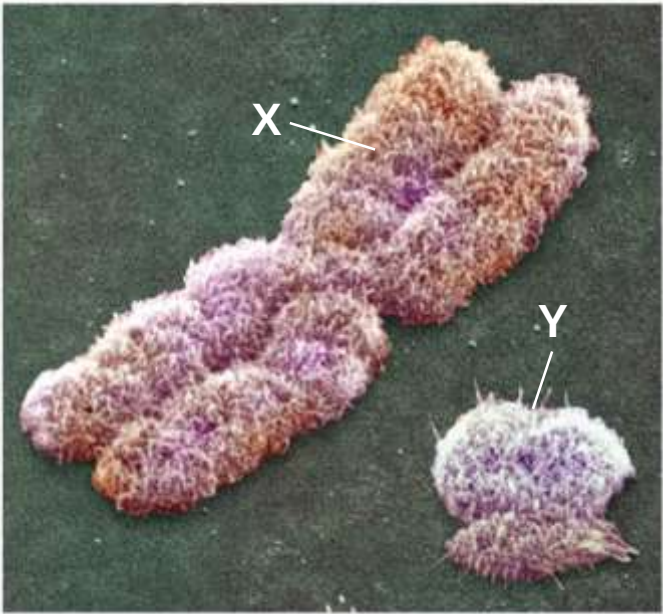
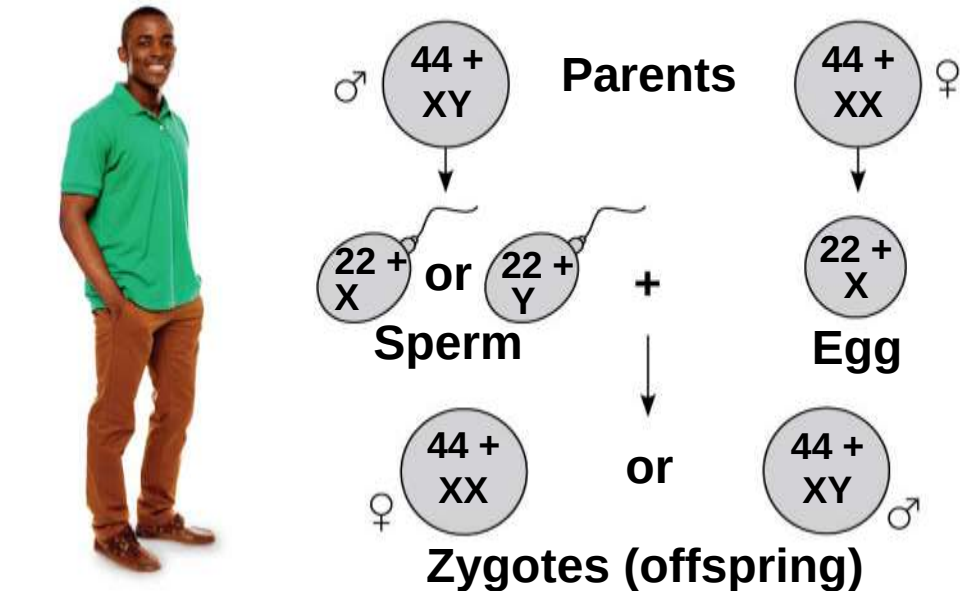
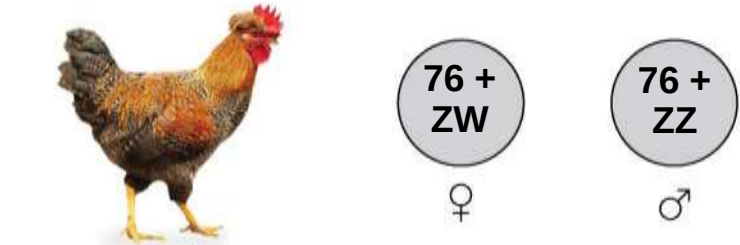


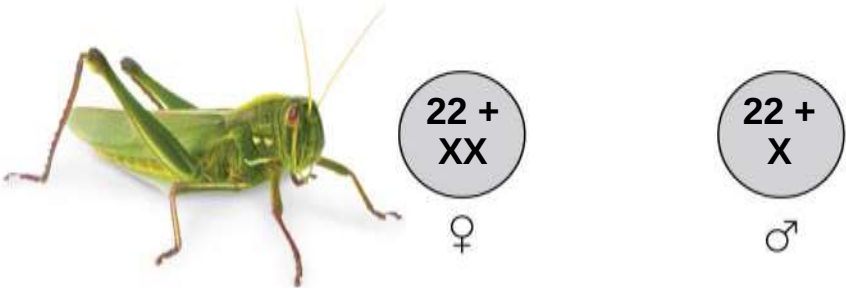
Figure 15.6



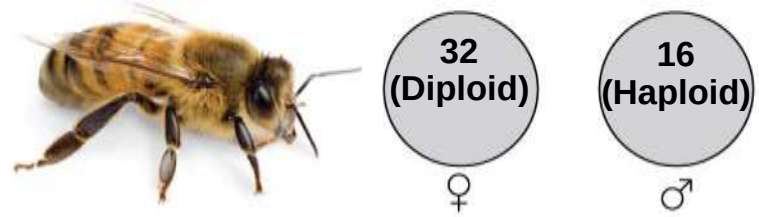
(a) The X-Y system



(c) The Z-W system

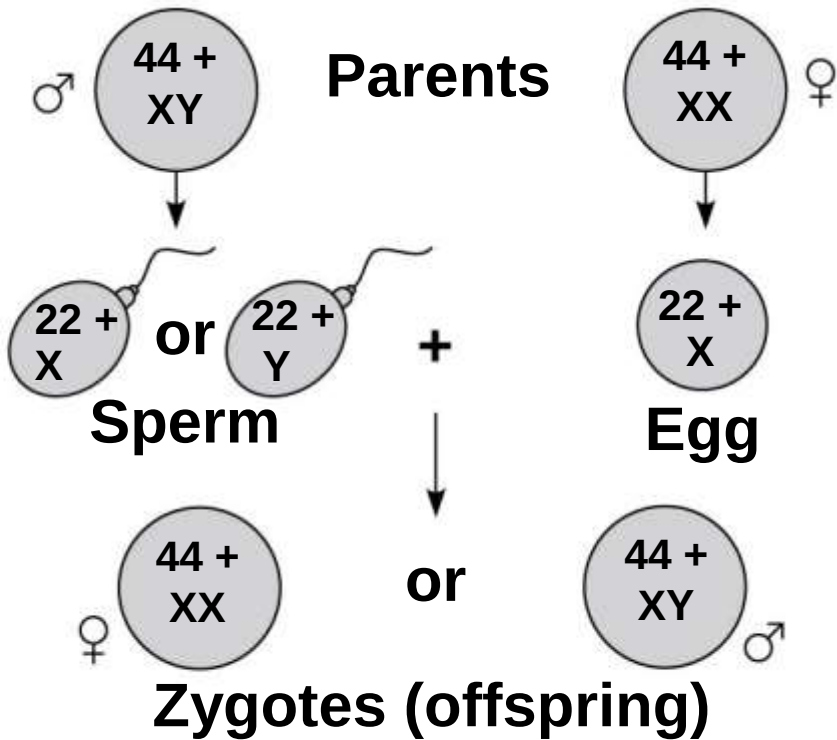


(b) The X-0 system



(d) The haplo-diploid system

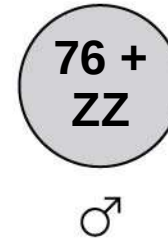
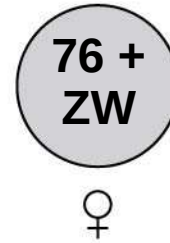
Figure 15.6a



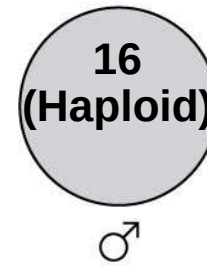
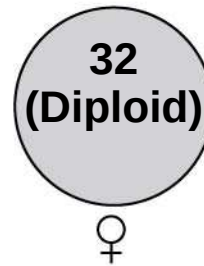
(a) The X-Y system



(b) The X-0 system



(c) The Z-W system



(d) The haplo-diploid system

- a) Short segments at the ends of the Y chromosomes are homologous with the X, allowing the two to behave like homologues during meiosis in males
- b) A gene on the Y chromosome called SRY (sex-determining region on the Y) is responsible for development of the testes in an embryo

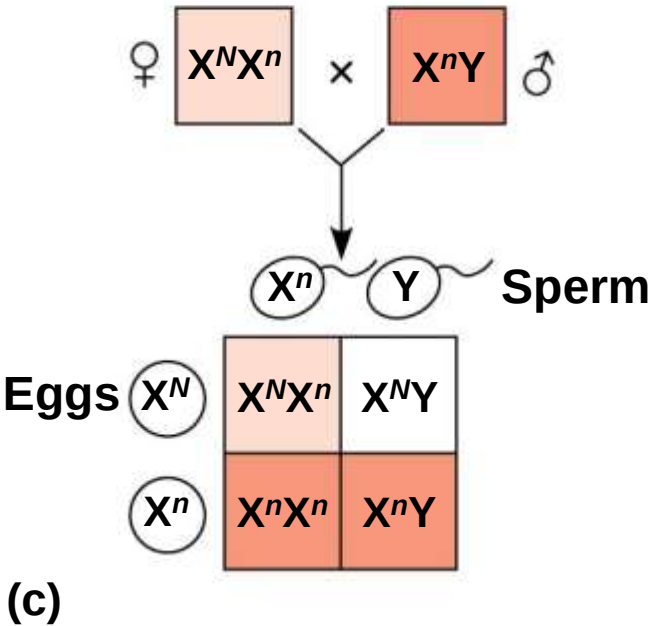
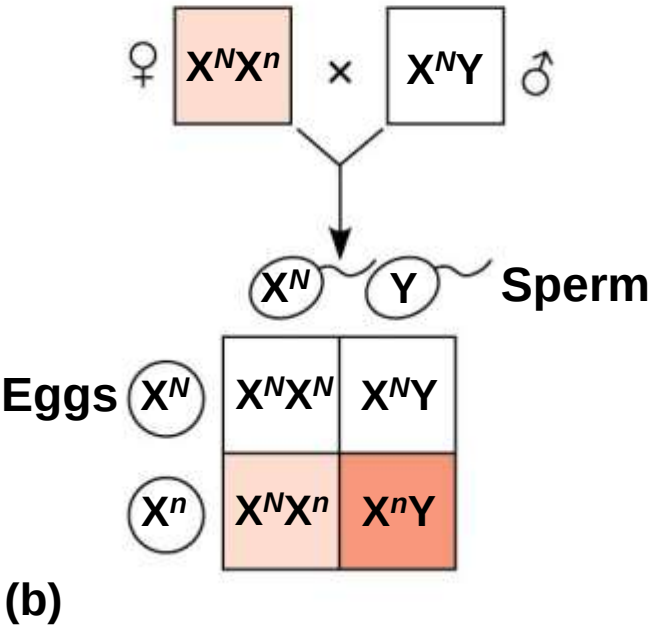
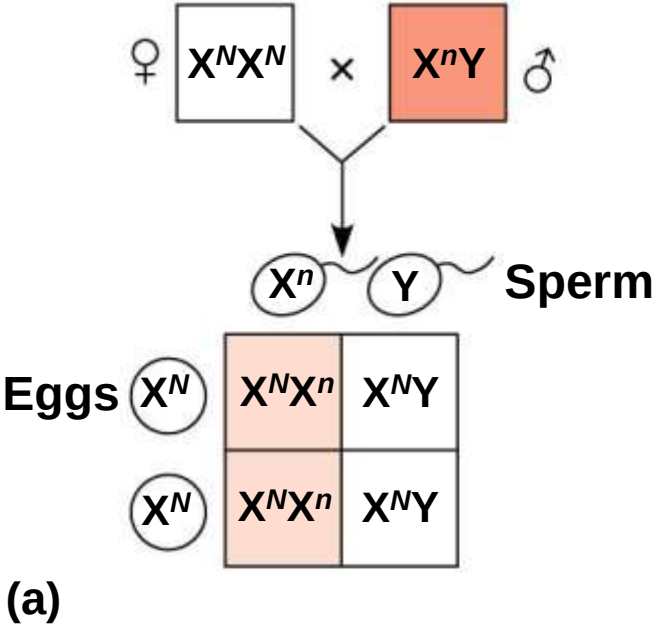
- a) A gene that is located on either sex chromosome is called a **sex-linked gene**
- b) Genes on the Y chromosome are called Y-linked genes; there are few of these
- c) Genes on the X chromosome are called **X-linked genes**

Inheritance of X-Linked Genes

- a) X chromosomes have genes for many characters unrelated to sex, whereas most Y-linked genes are related to sex determination

- a) X-linked genes follow specific patterns of inheritance
- b) For a recessive X-linked trait to be expressed
 - a) A female needs two copies of the allele (homozygous)
 - b) A male needs only one copy of the allele (hemizygous)
- c) X-linked recessive disorders are much more common in males than in females

Figure 15.7



a) Some disorders caused by recessive alleles on the X chromosome in humans

a) Color blindness (mostly X-linked)

b) Duchenne muscular dystrophy

c) Hemophilia

X Inactivation in Female Mammals

- a) In mammalian females, one of the two X chromosomes in each cell is randomly inactivated during embryonic development
- b) The inactive X condenses into a **Barr body**
- c) If a female is heterozygous for a particular gene located on the X chromosome, she will be a mosaic for that character

Figure 15.8

Early embryo:

X chromosomes

Allele for orange fur
Allele for black fur

Cell division and X chromosome inactivation

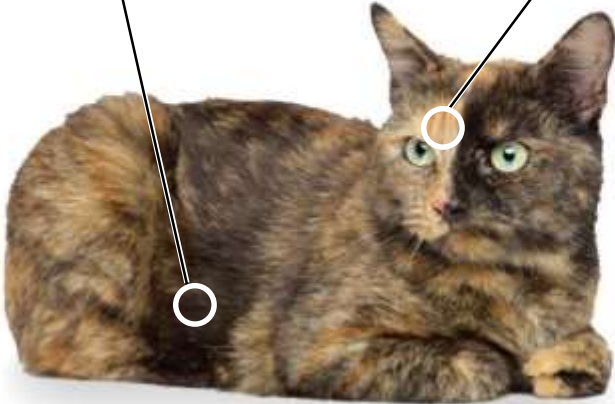
Two cell populations in adult cat:
Active X

Inactive X

Active X

Black fur

Orange fur



Concept 15.3: Linked genes tend to be inherited together because they are located near each other on the same chromosome

- a) Each chromosome has hundreds or thousands of genes (except the Y chromosome)
- b) Genes located on the same chromosome that tend to be inherited together are called **linked genes**

How Linkage Affects Inheritance

- a) Morgan did other experiments with fruit flies to see how linkage affects inheritance of two characters
- b) Morgan crossed flies that differed in traits of body color and wing size

Figure 15.9

Experiment

P Generation

(homozygous)

Wild type (gray
body, normal wings)

$b^+ b^+ vg^+ vg^+$



x



Double mutant
(black body, vestigial wings)

$b b vg vg$

F₁ dihybrid testcross

Wild-type F₁ dihybrid
(gray body, normal wings)

$b^+ b vg^+ vg$



x



Homozygous
recessive (black
body, vestigial wings)

$b b vg vg$

Testcross
offspring

Eggs

$b^+ vg^+$

$b vg$

$b^+ vg$

$b vg^+$

$b vg$
Sperm

Wild type (gray-normal)	Black- vestigial	Gray- vestigial	Black- normal
$b^+ b vg^+ vg$	$b b vg vg$	$b^+ b vg vg$	$b b vg^+ vg$

PREDICTED RATIOS

Genes on different
chromosomes:

1 : 1 : 1 : 1

Genes on the same
chromosome:

1 : 1 : 0 : 0

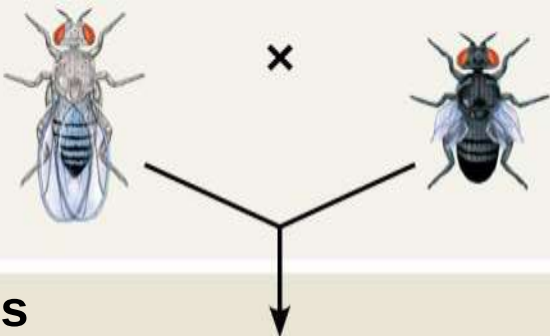
Results

965 : 944 : 206 : 185

Figure 15.9a

Experiment

P Generation
(homozygous)
Wild type (gray
body, normal
wings)
b⁺ b⁺ vg⁺ vg⁺



Double mutant
(black body, vestigial
wings)
b b vg vg

F₁ dihybrid testcross
Wild-type F₁ dihybrid
(gray body, normal
wings)
b⁺ b vg⁺ vg



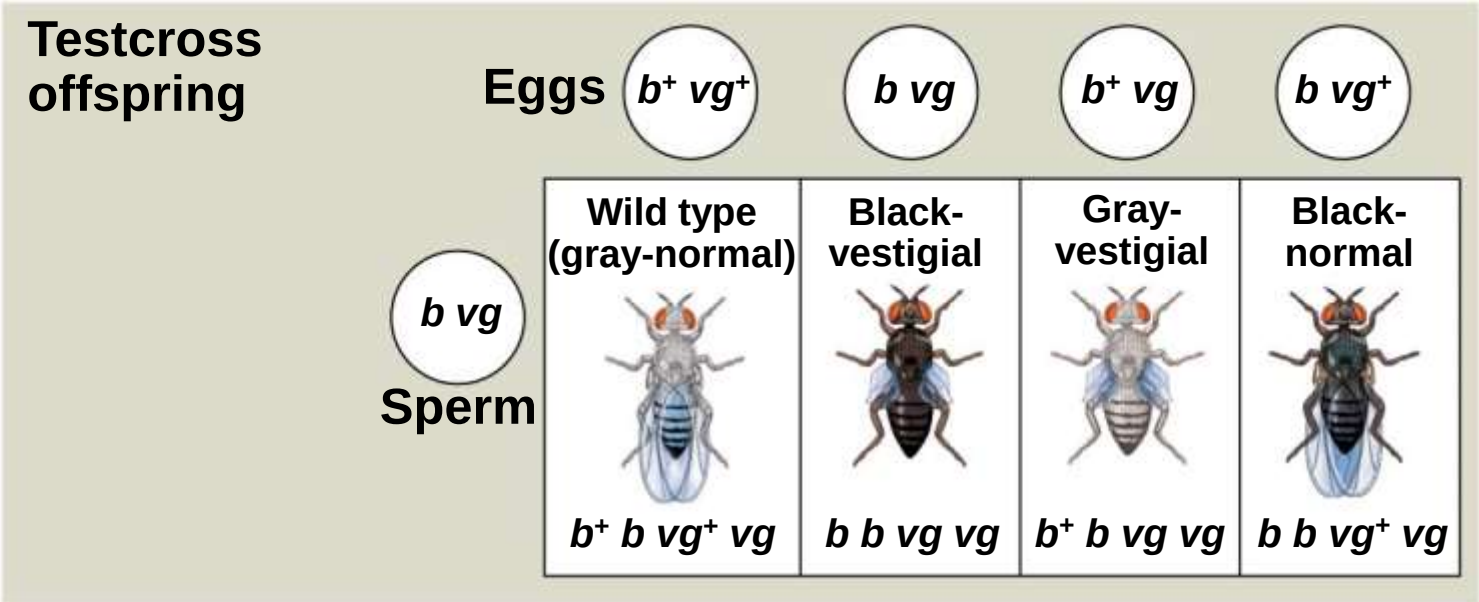
x



♂

**Homozygous
recessive** (black
body, vestigial
wings)
b b vg vg

Experiment

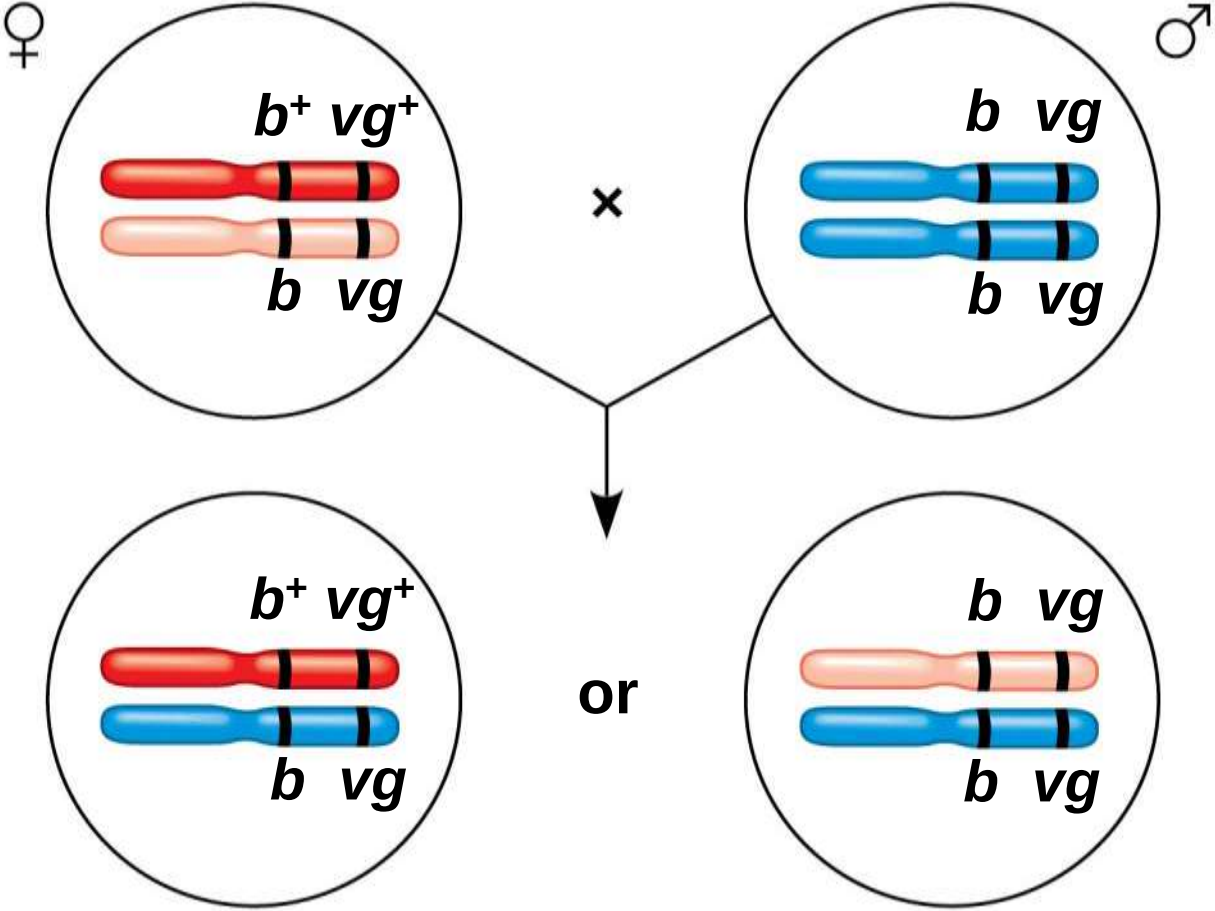


PREDICTED RATIOS

Genes on different chromosomes:	1	:	1	:	1	:	1
Genes on the same chromosome:	1	:	1	:	0	:	0
Results	965	:	944	:	206	:	185

- a) Morgan found that body color and wing size are usually inherited together in specific combinations (parental phenotypes)
- b) He noted that these genes do not assort independently, and reasoned that they were on the same chromosome

**F₁ dihybrid female
and homozygous
recessive male
in testcross**



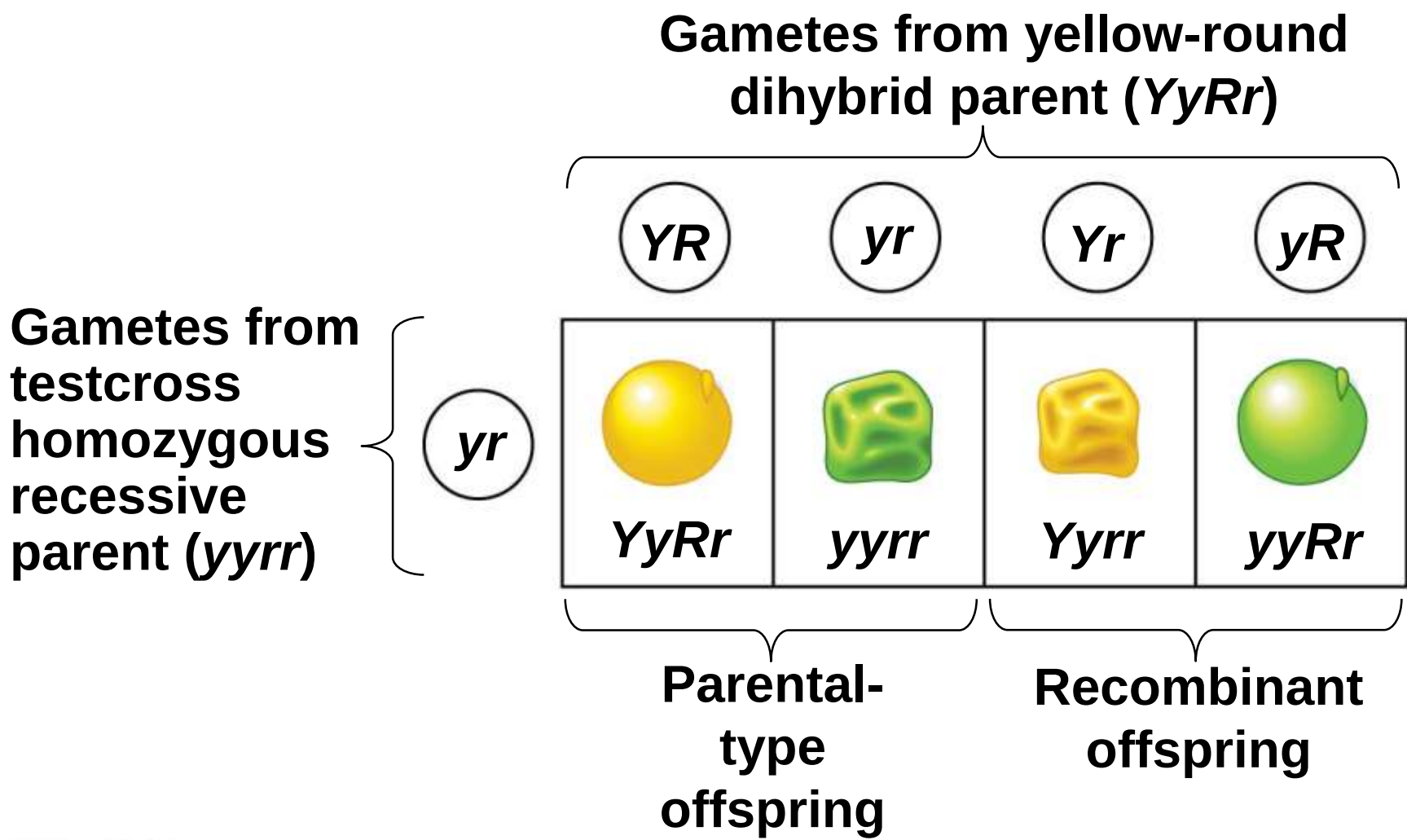
- a) However, nonparental phenotypes were also produced
- b) Understanding this result involves exploring **genetic recombination**, the production of offspring with combinations of traits differing from either parent

Genetic Recombination and Linkage

a) The genetic findings of Mendel and Morgan relate to the chromosomal basis of recombination

Recombination of Unlinked Genes: Independent Assortment of Chromosomes

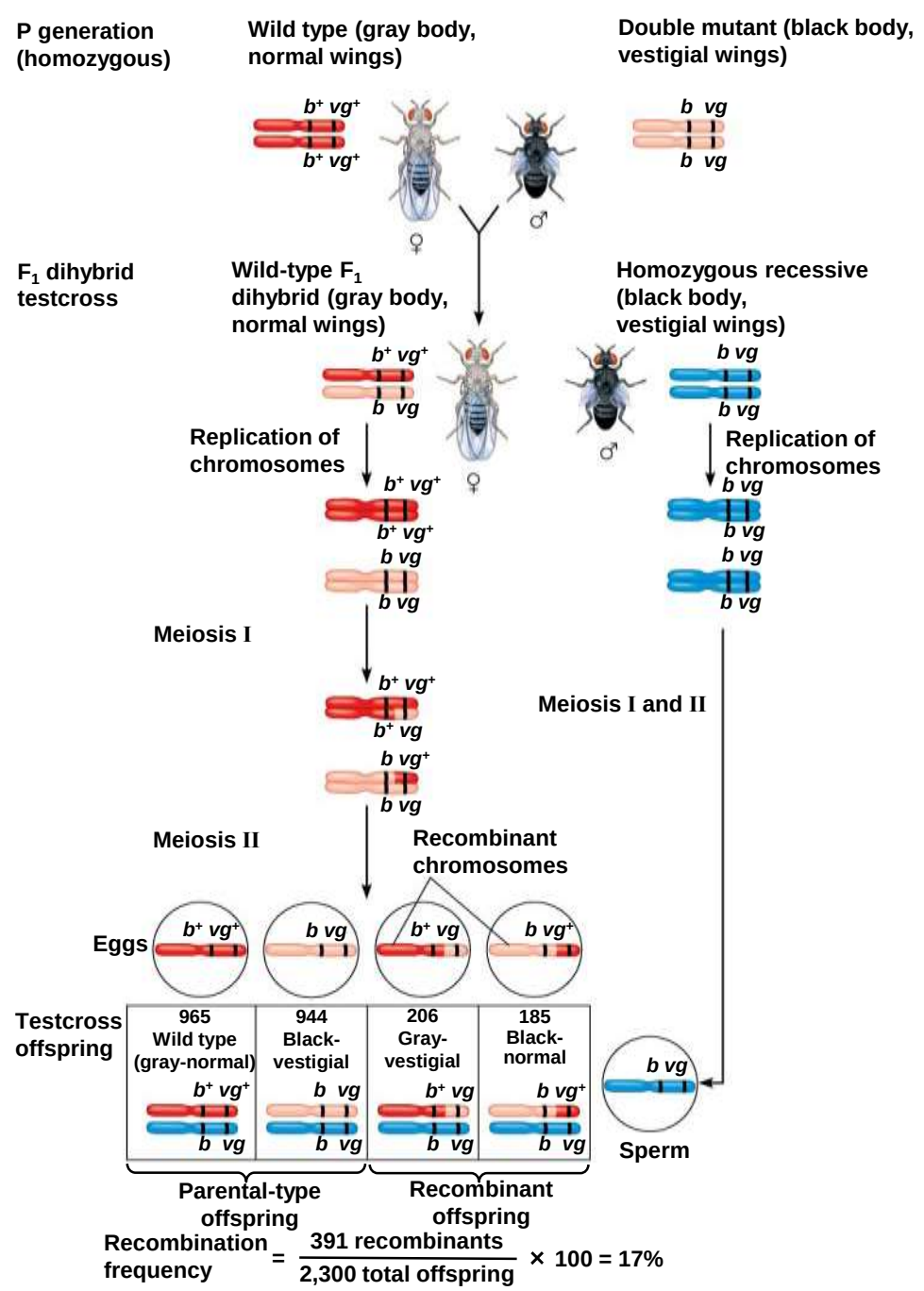
- a) Offspring with a phenotype matching one of the parental phenotypes are called **parental types**
- b) Offspring with nonparental phenotypes (new combinations of traits) are called **recombinant types**, or **recombinants**
- c) A 50% frequency of recombination is observed for any two genes on different chromosomes



Recombination of Linked Genes: Crossing Over

- a) Morgan discovered that genes can be linked, but the linkage was incomplete, because some recombinant phenotypes were observed
- b) He proposed that some process must occasionally break the physical connection between genes on the same chromosome
- c) That mechanism was the **crossing over** of homologous chromosomes

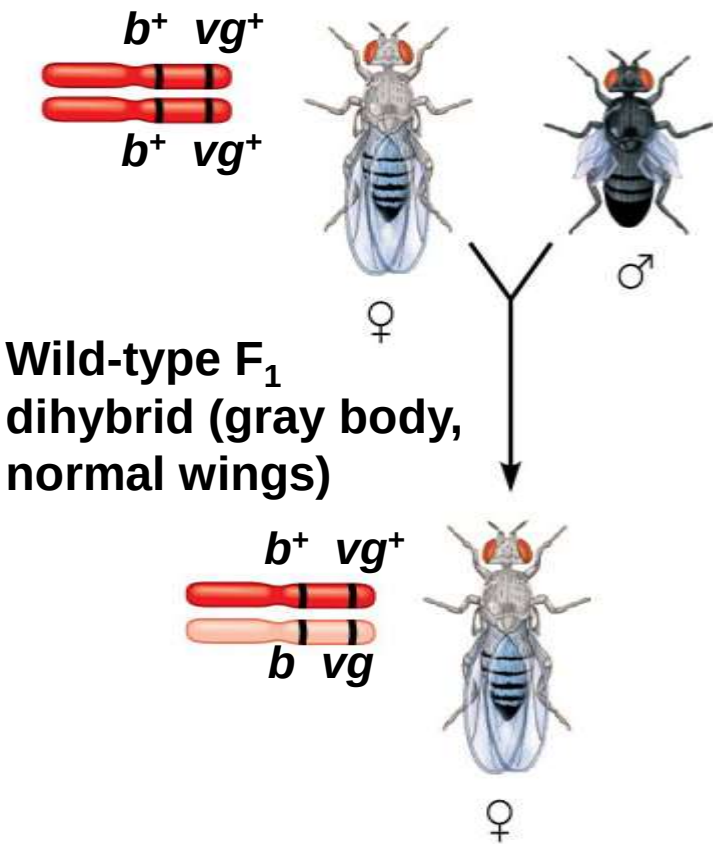
Figure 15.10



P generation (homozygous)

Wild type (gray body, normal wings)

Double mutant (black body, vestigial wings)



Wild-type F₁ dihybrid (gray body, normal wings)

Figure 15.10b

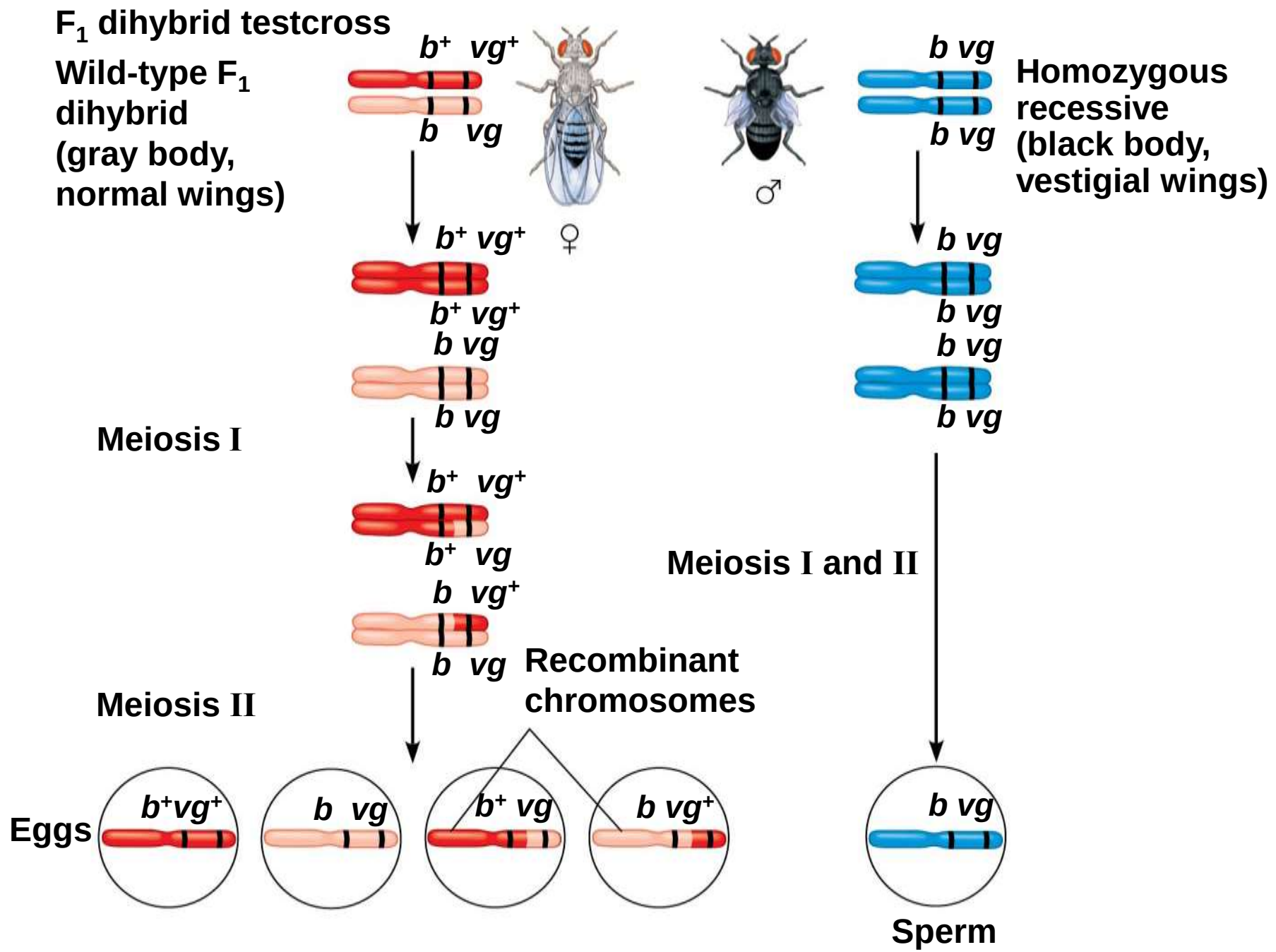
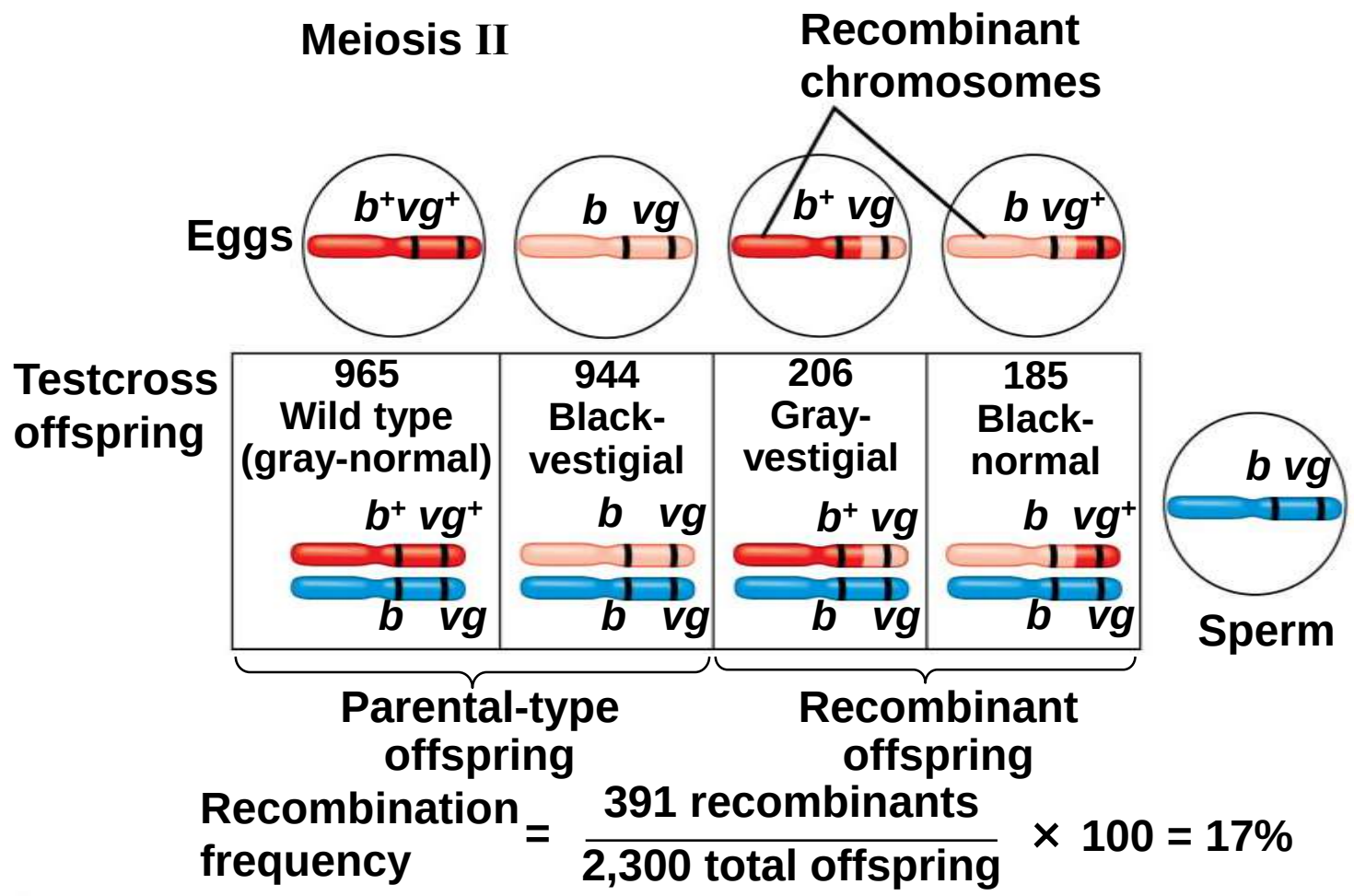
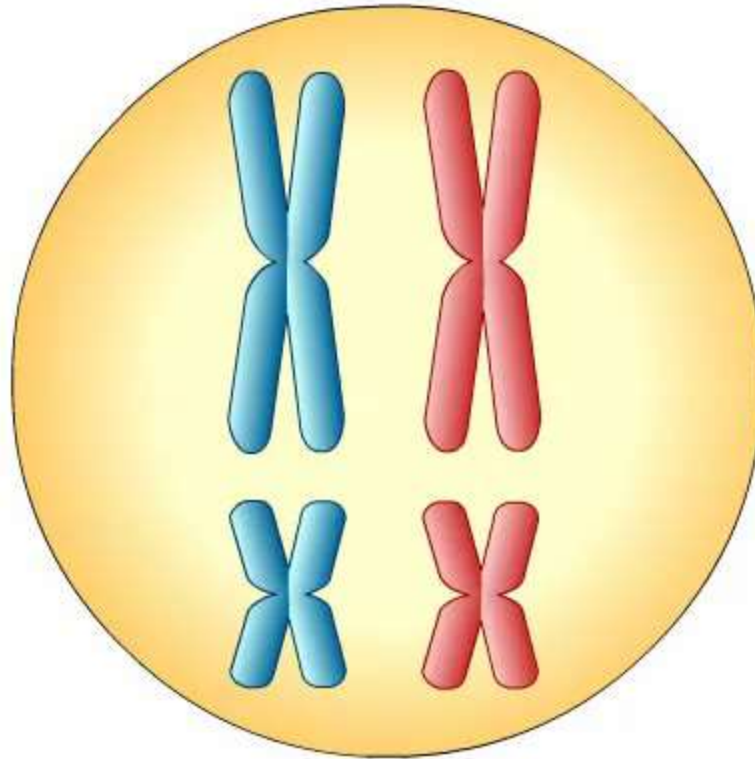


Figure 15.10c



Animation: Crossing Over



New Combinations of Alleles: Variation for Natural Selection

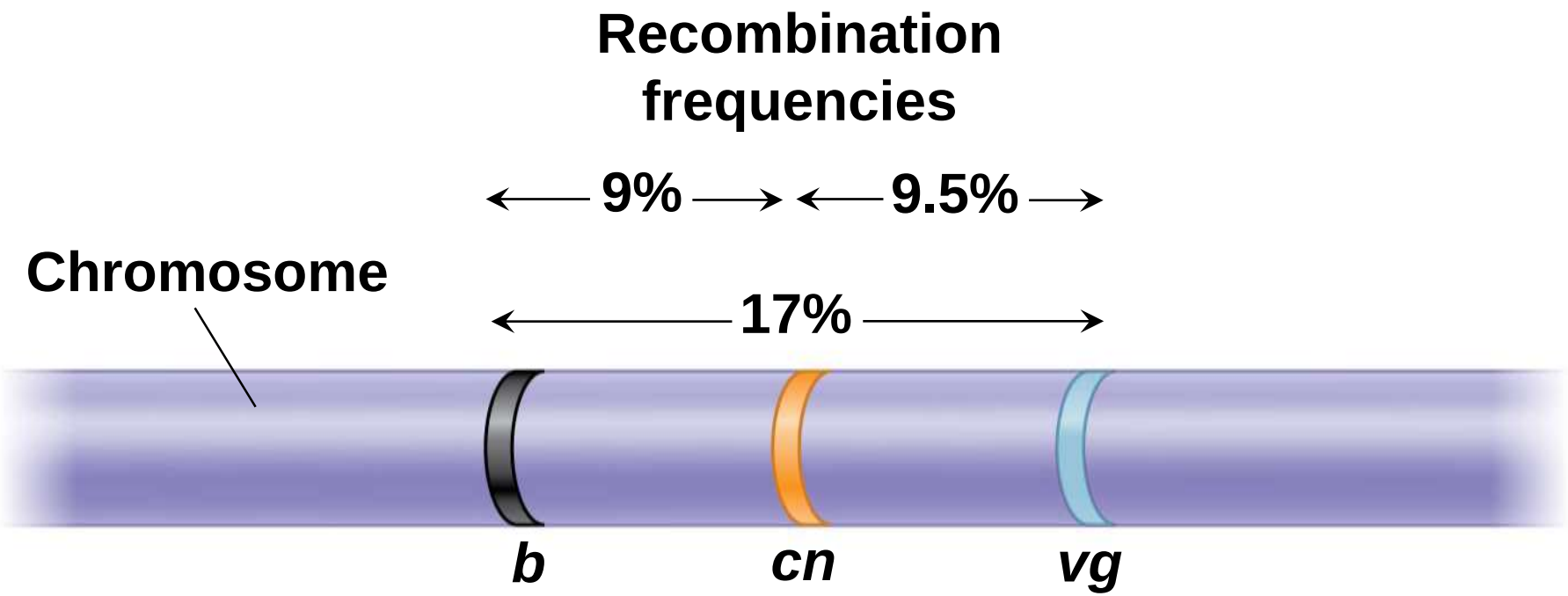
- a) Recombinant chromosomes bring alleles together in new combinations in gametes
- b) Random fertilization increases even further the number of variant combinations that can be produced
- c) This abundance of genetic variation is the raw material upon which natural selection works

Mapping the Distance Between Genes Using Recombination Data: *Scientific Inquiry*

- a) Alfred Sturtevant, one of Morgan's students, constructed a **genetic map**, an ordered list of the genetic loci along a particular chromosome
- b) Sturtevant predicted that *the farther apart two genes are, the higher the probability that a crossover will occur between them and therefore the higher the recombination frequency*

- a) A **linkage map** is a genetic map of a chromosome based on recombination frequencies
- b) Distances between genes can be expressed as **map units**; one map unit, or centimorgan, represents a 1% recombination frequency
- c) Map units indicate relative distance and order, not precise locations of genes

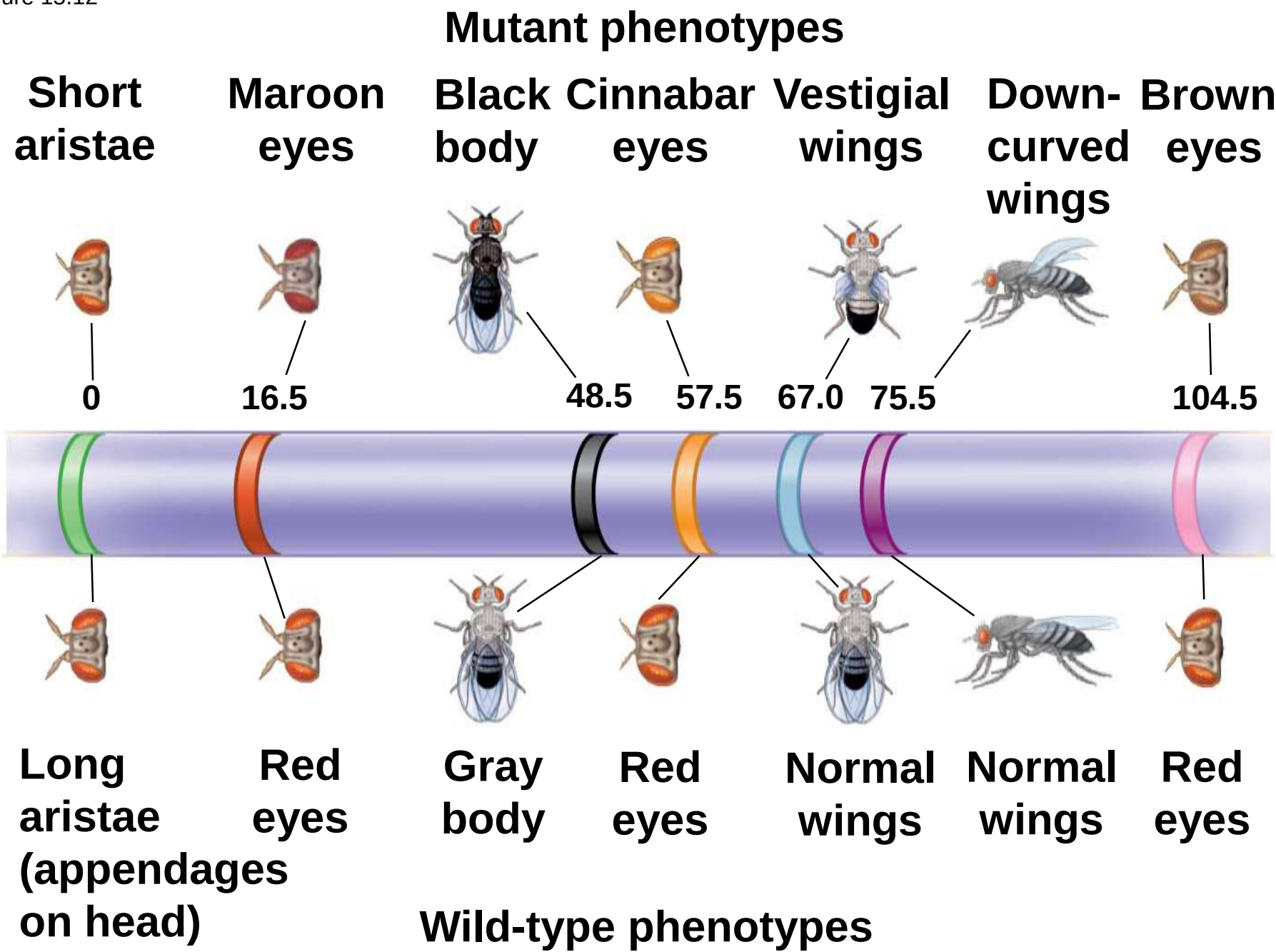
Results



- a) Genes that are far apart on the same chromosome can have a recombination frequency near 50%
- b) Such genes are physically linked, but genetically unlinked, and behave as if found on different chromosomes

- a) Sturtevant used recombination frequencies to make linkage maps of fruit fly genes
- b) He and his colleagues found that the genes clustered into four groups of linked genes (linkage groups)
- c) The linkage maps, combined with the fact that there are four chromosomes in *Drosophila*, provided additional evidence that genes are located on chromosomes

Figure 15.12



Concept 15.4: Alterations of chromosome number or structure cause some genetic disorders

- a) Large-scale chromosomal alterations in humans and other mammals often lead to spontaneous abortions (miscarriages) or cause a variety of developmental disorders
- b) Plants tolerate such genetic changes better than animals do

Abnormal Chromosome Number

- a) In **nondisjunction**, pairs of homologous chromosomes do not separate normally during meiosis
- b) As a result, one gamete receives two of the same type of chromosome, and another gamete receives no copy

Figure 15.13-1



Figure 15.13-2

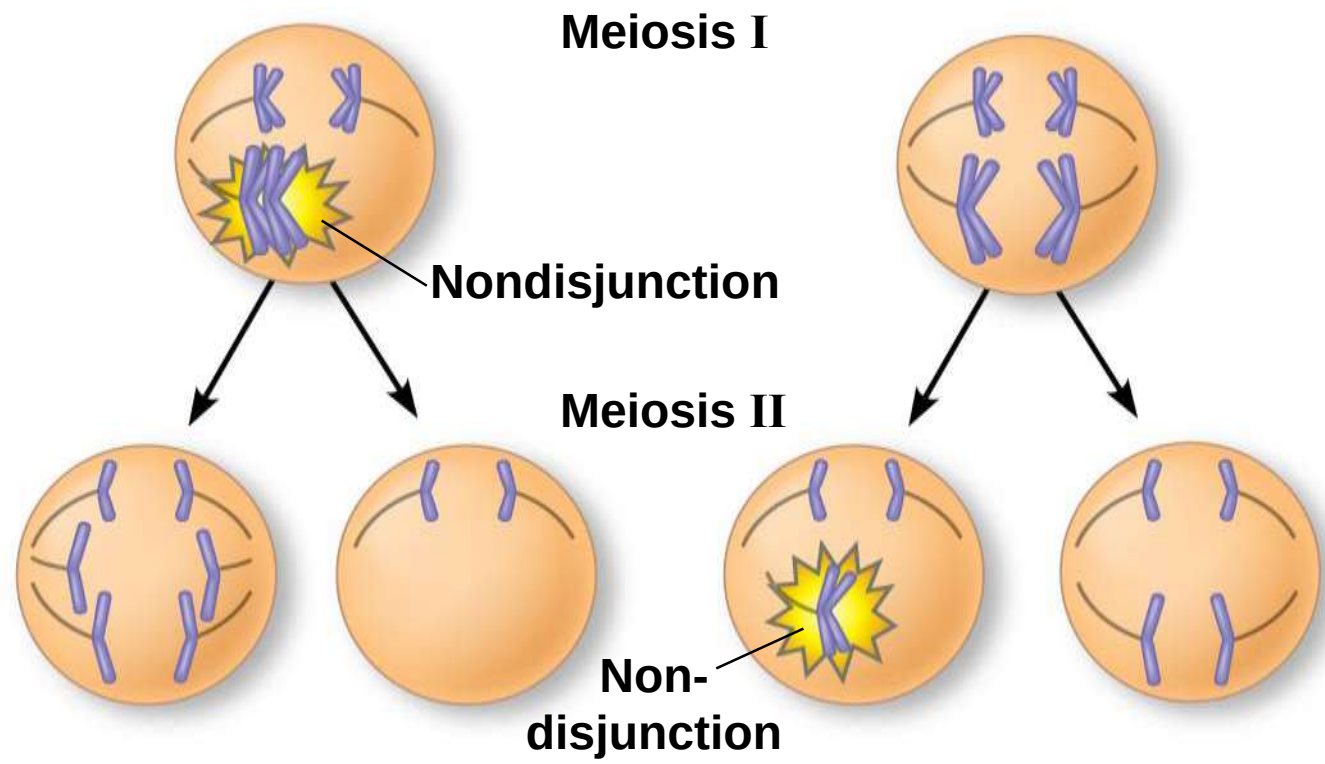
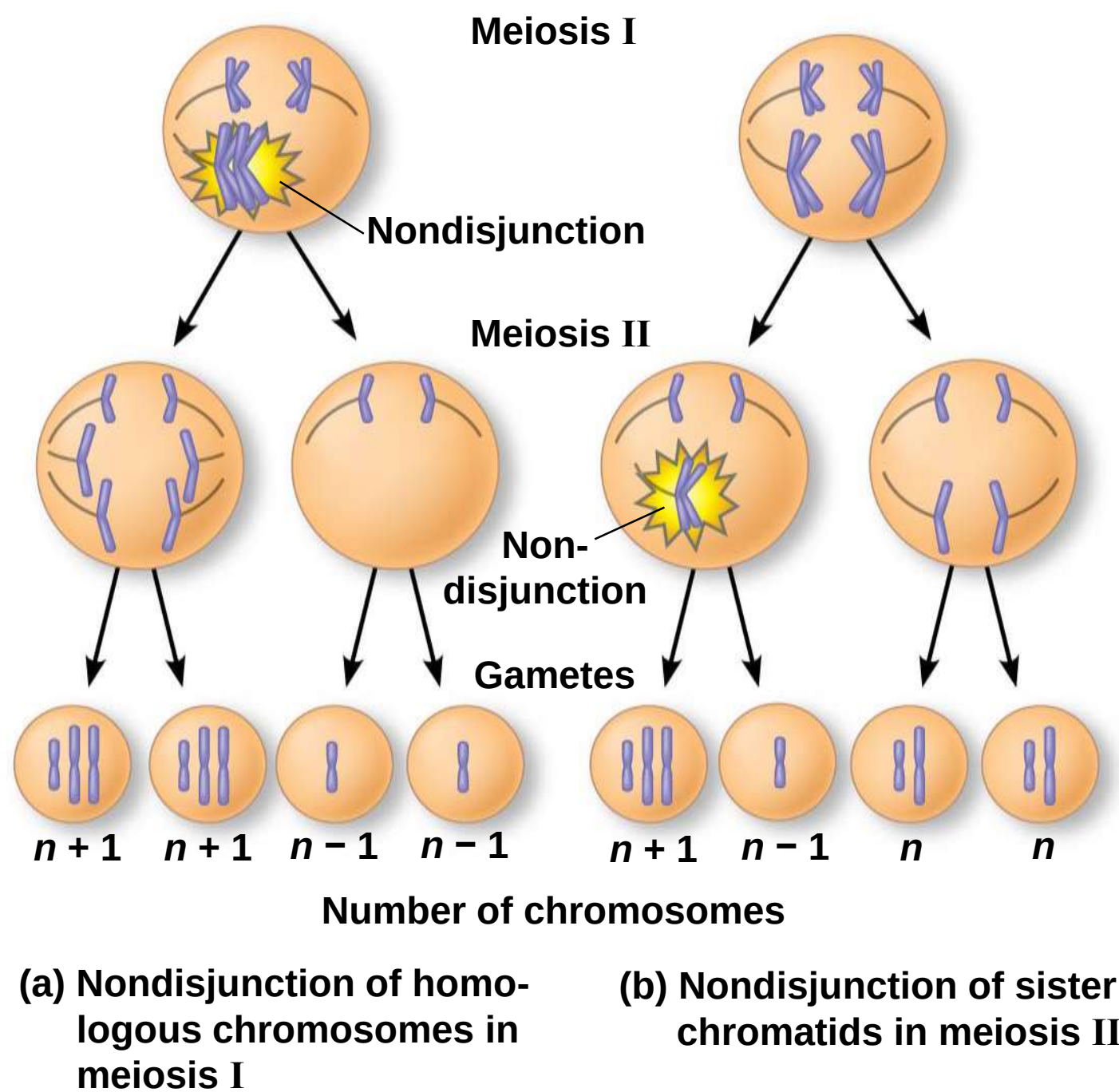


Figure 15.13-3



Video: Nondisjunction in Mitosis



- a) **Aneuploidy** results from the fertilization of gametes in which nondisjunction occurred
- b) Offspring with this condition have an abnormal number of a particular chromosome

- a) A **monosomic** zygote has only one copy of a particular chromosome
- b) A **trisomic** zygote has three copies of a particular chromosome

a) Polyploidy is a condition in which an organism has more than two complete sets of chromosomes

a) Triploidy ($3n$) is three sets of chromosomes

b) Tetraploidy ($4n$) is four sets of chromosomes

b) Polyploidy is common in plants, but not animals

c) Polyploids are more normal in appearance than aneuploids

Alterations of Chromosome Structure

a) Breakage of a chromosome can lead to four types of changes in chromosome structure

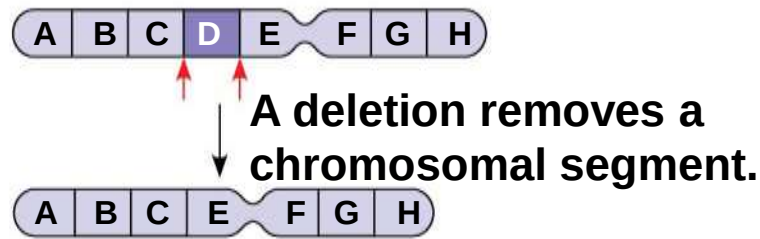
a) Deletion removes a chromosomal segment

b) Duplication repeats a segment

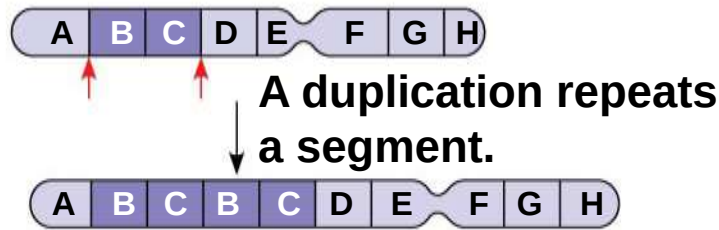
c) Inversion reverses orientation of a segment within a chromosome

d) Translocation moves a segment from one chromosome to another

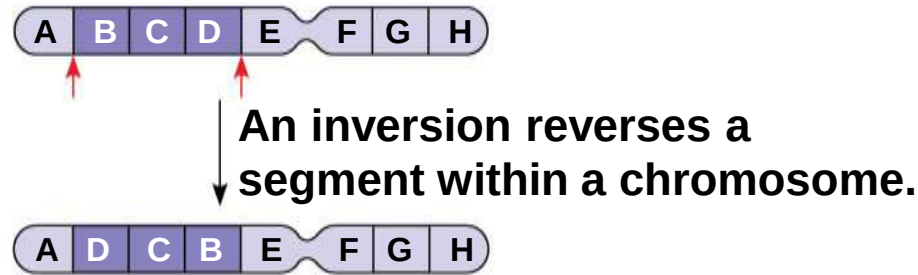
(a) Deletion



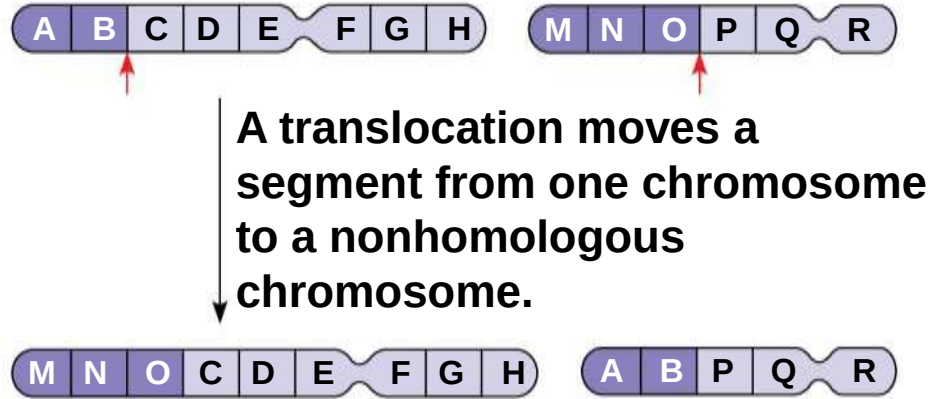
(b) Duplication



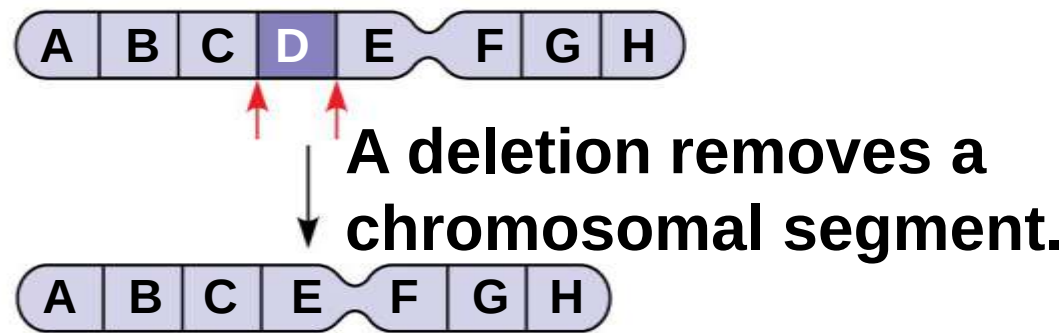
(c) Inversion



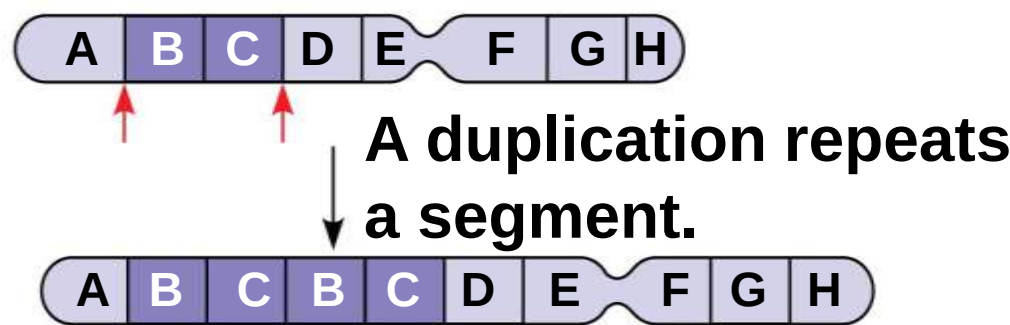
(d) Translocation



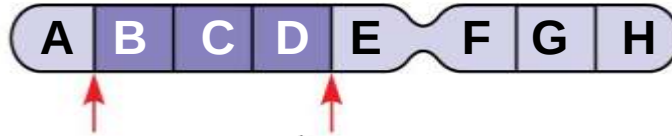
(a) Deletion



(b) Duplication



(c) Inversion



An inversion reverses a segment within a chromosome.



(d) Translocation



A translocation moves a segment from one chromosome to a nonhomologous chromosome.



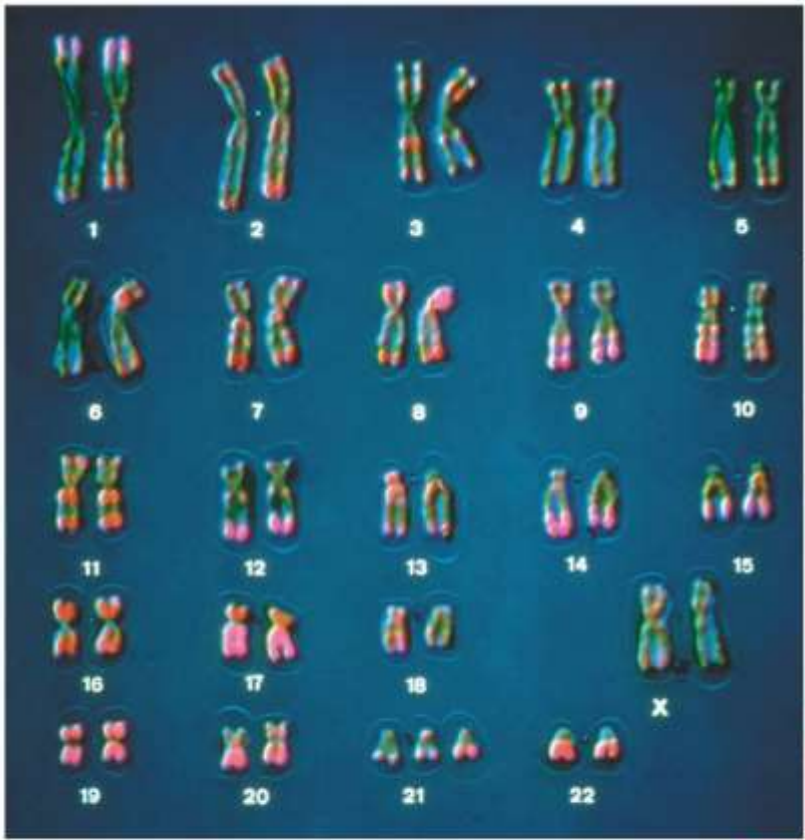
Human Disorders Due to Chromosomal Alterations

- a) Alterations of chromosome number and structure are associated with some serious disorders
- b) Some types of aneuploidy appear to upset the genetic balance less than others, resulting in individuals surviving to birth and beyond
- c) These surviving individuals have a set of symptoms, or syndrome, characteristic of the type of aneuploidy

Down Syndrome (Trisomy 21)

- a) **Down syndrome** is an aneuploid condition that results from three copies of chromosome 21
- b) It affects about one out of every 700 children born in the United States
- c) The frequency of Down syndrome increases with the age of the mother, a correlation that has not been explained

Figure 15.15



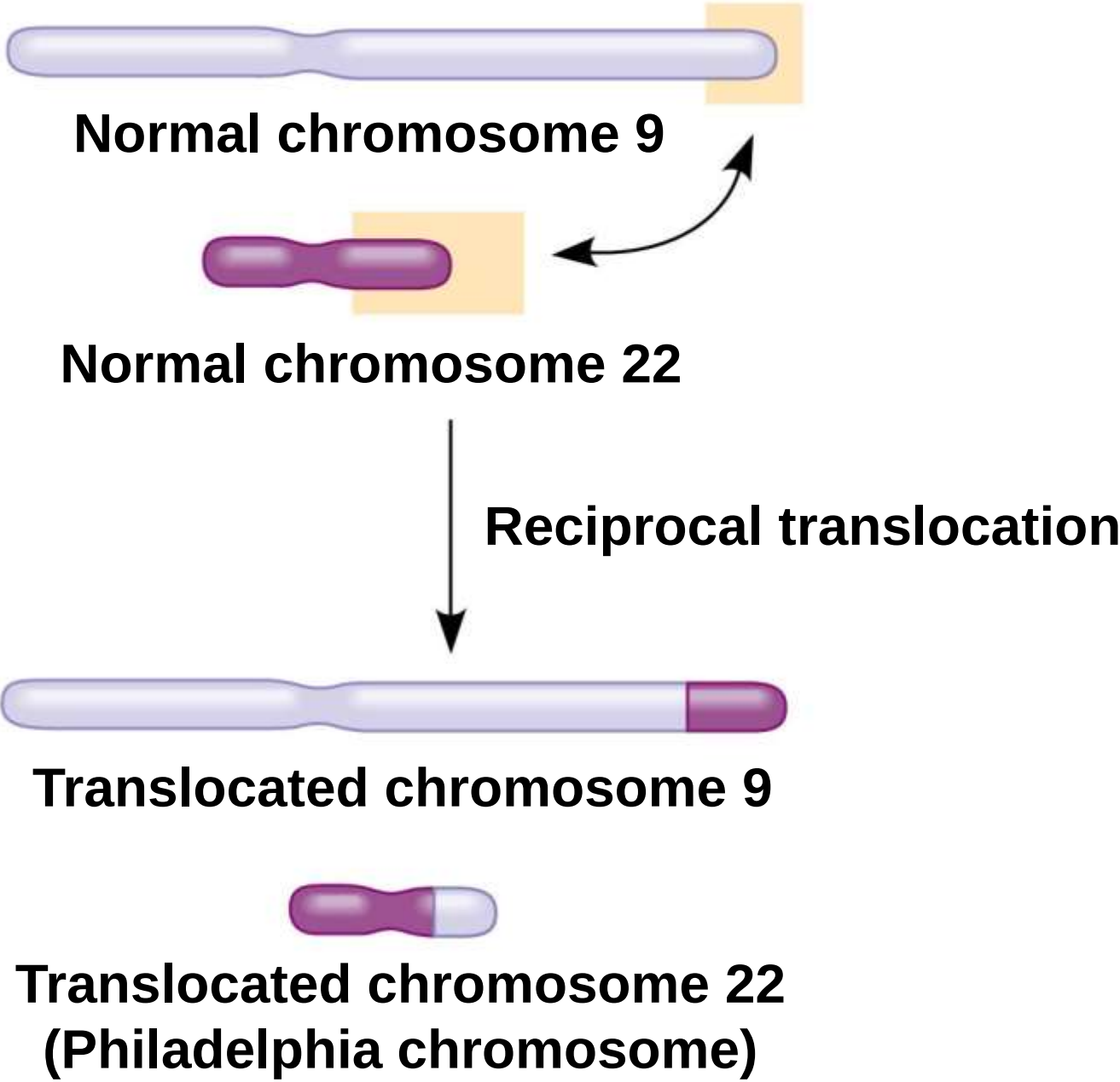
Aneuploidy of Sex Chromosomes

- a) Nondisjunction of sex chromosomes produces a variety of aneuploid conditions
- b) XXX females are healthy, with no unusual physical features
- c) Klinefelter syndrome is the result of an extra chromosome in a male, producing XXY individuals
- d) Monosomy X, called *Turner syndrome*, produces X0 females, who are sterile; it is the only known viable monosomy in humans

Disorders Caused by Structurally Altered Chromosomes

- a) The syndrome *cri du chat* (“cry of the cat”), results from a specific deletion in chromosome 5
- b) A child born with this syndrome is severely intellectually disabled and has a catlike cry; individuals usually die in infancy or early childhood
- c) Certain cancers, including *chronic myelogenous leukemia* (CML), are caused by translocations of chromosomes

Figure 15.16

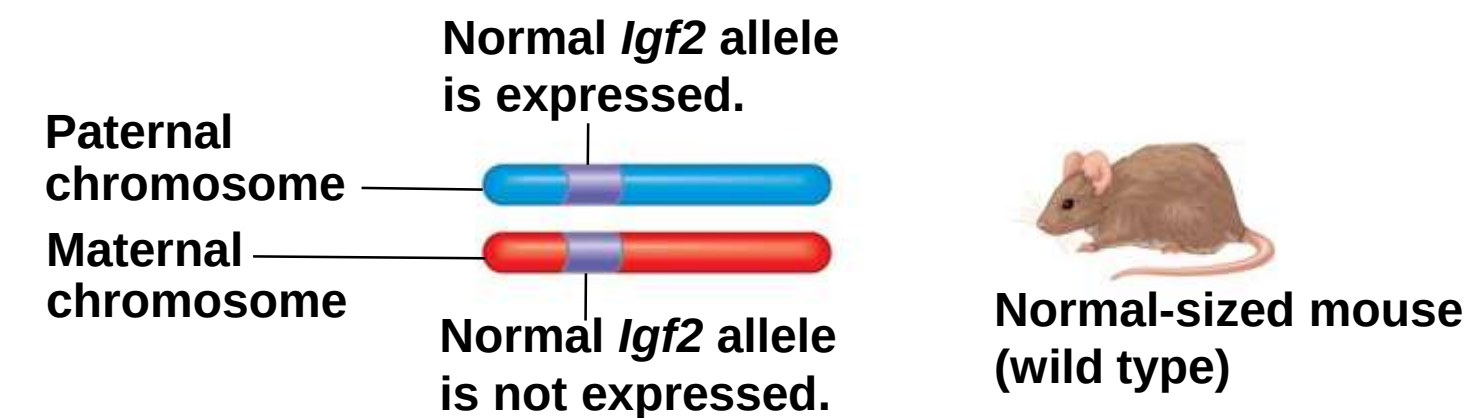


Concept 15.5: Some inheritance patterns are exceptions to standard Mendelian inheritance

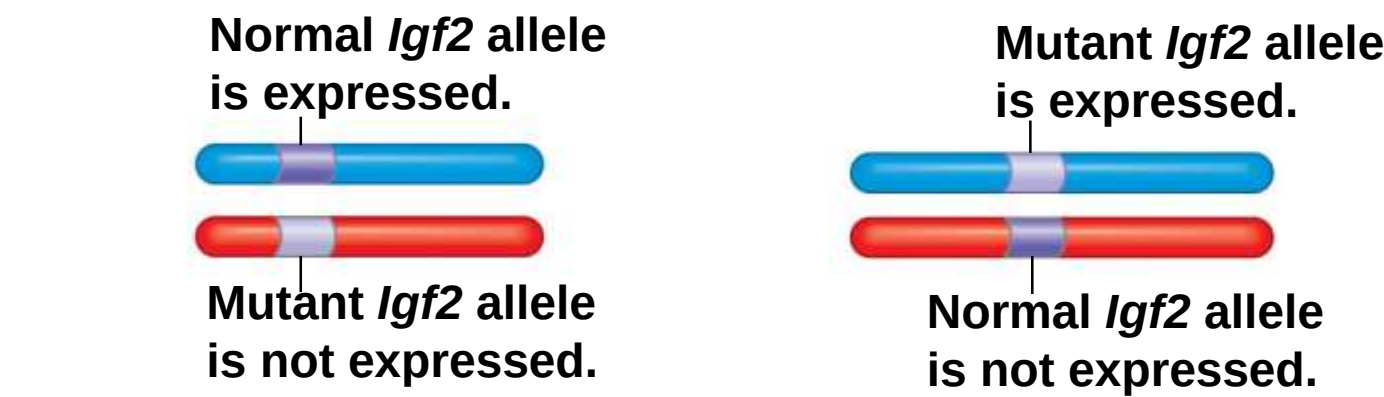
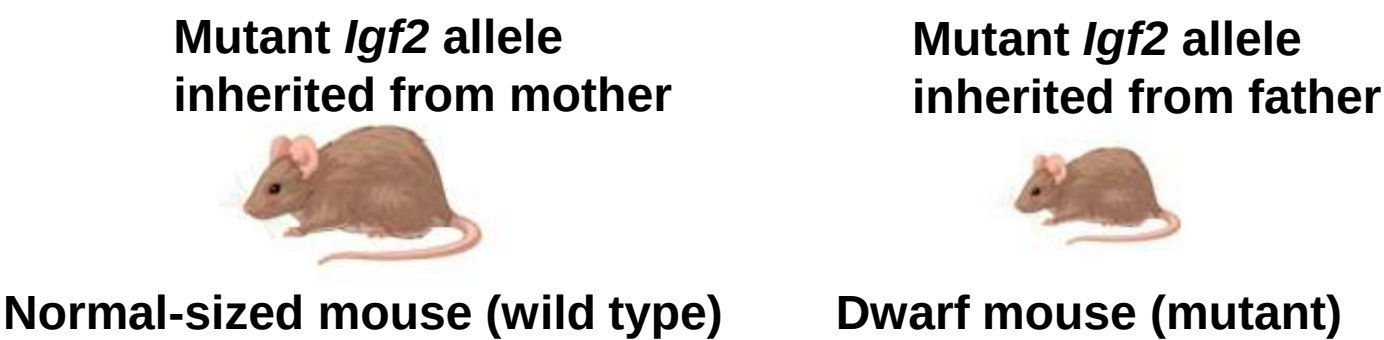
- a) There are two normal exceptions to Mendelian genetics
- b) One exception involves genes located in the nucleus, and the other exception involves genes located outside the nucleus
- c) In both cases, the sex of the parent contributing an allele is a factor in the pattern of inheritance

Genomic Imprinting

- a) For a few mammalian traits, the phenotype depends on which parent passed along the alleles for those traits
- b) Such variation in phenotype is called **genomic imprinting**
- c) Genomic imprinting involves the silencing of certain genes depending on which parent passes them on



(a) Homozygote



(b) Heterozygotes

- a) It appears that imprinting is the result of the methylation (addition of —CH_3) of cytosine nucleotides
- b) Genomic imprinting is thought to affect only a small fraction of mammalian genes
- c) Most imprinted genes are critical for embryonic development

Inheritance of Organelle Genes

- a) Extranuclear genes (or cytoplasmic genes) are found in organelles in the cytoplasm
- b) Mitochondria, chloroplasts, and other plant plastids carry small circular DNA molecules
- c) Extranuclear genes are inherited maternally because the zygote's cytoplasm comes from the egg
- d) The first evidence of extranuclear genes came from studies on the inheritance of yellow or white patches on leaves of an otherwise green plant

Figure 15.18



- a) Some defects in mitochondrial genes prevent cells from making enough ATP and result in diseases that affect the muscular and nervous systems
 - a) For example, mitochondrial myopathy and Leber's hereditary optic neuropathy